

Instrumental Variable Approaches

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2026-06-07

The endogeneity problem

We want the **causal** effect β of a treatment x on an outcome y :

$$y_i = \alpha + \beta x_i + \varepsilon_i.$$

OLS recovers β only if x_i is uncorrelated with everything else in ε_i . It **fails** whenever $\text{Cov}(x_i, \varepsilon_i) \neq 0$:

- **Omitted variables** — a confounder drives both x and y .
- **Reverse causality / simultaneity** — y also affects x (supply *and* demand).
- **Measurement error** in x (attenuation).

In that case

$$\text{plim } \hat{\beta}_{OLS} = \beta + \frac{\text{Cov}(x_i, \varepsilon_i)}{\text{Var}(x_i)} \neq \beta.$$

Where we are going today:

- ➊ **Instrumental variables** — the two conditions, the estimator, and diagnostics.
- ➋ **A detour: Generalized Method of Moments (GMM)** — for your culture, not your toolkit.
- ➌ **The shift-share (Bartik) instrument** — what it is and how it is built.
- ➍ **Identification “from the shares”** — Goldsmith-Pinkham et al. (2020).
- ➎ **Identification “from the shifts”** — Borusyak et al. (2022).
- ➏ **Inference** — Adão et al. (2019).
- ➐ **A practical guide** — which assumption fits your design?
- ➑ **Applied session** — estimating a Bartik design in Python.

Two readings underpin the whole lecture: Goldsmith-Pinkham et al. (2020) on shares, Borusyak et al. (2022) on shifts.

Section 1

Instrumental Variables

The IV solution: two conditions

An **instrument** z is a variable that moves x but is otherwise unrelated to the outcome. It must satisfy two conditions:

Relevance (testable)

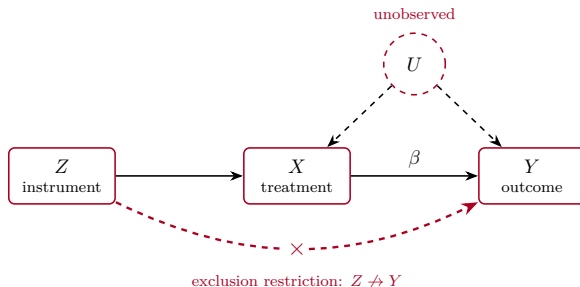
$\text{Cov}(z_i, x_i) \neq 0$ — the instrument actually shifts the treatment. Checked with the *first stage*.

Exogeneity / exclusion (not testable)

$\text{Cov}(z_i, \varepsilon_i) = 0$ — equivalently the **orthogonality (moment) condition** $\mathbb{E}[z_i \varepsilon_i] = 0$. The instrument affects y *only* through x , not through any other channel. Defended with economic argument, not a test.

The art of IV is almost entirely about **defending exogeneity**.

The logic in one picture



- **Relevance:** $Z \rightarrow X$ — the instrument moves the treatment.
- **Confounding:** U (unobserved) drives both X and Y — this is *why OLS is biased*.
- **Exclusion:** there is **no** arrow $Z \rightarrow Y$ (and none $U \rightarrow Z$) — so Z touches Y *only* through X . IV uses just the Z -driven part of X , which is clean of U .

The IV / Wald estimator

With a single instrument, take the covariance of the structural equation with z :

$$\text{Cov}(z, y) = \beta \text{Cov}(z, x) + \underbrace{\text{Cov}(z, \varepsilon)}_{=0}.$$

Solving for β gives the **IV estimator**:

$$\hat{\beta}_{IV} = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)}$$

Intuition: scale the instrument's effect on the outcome by its effect on the treatment. If z is binary, this is the **Wald estimator** — a ratio of differences in means.

The numerator and denominator are each just **simple regressions on z** — they have names, which we introduce next.

First stage, reduced form, and the IV ratio

The IV estimate is built from **two ordinary regressions on the instrument**:

First stage — does z move the treatment?

Regress the treatment on the instrument: $x_i = \pi z_i + \nu_i$.

The slope $\hat{\pi}$ measures **relevance**.

Reduced form — does z move the outcome?

Regress the outcome on the instrument: $y_i = \rho z_i + \eta_i$.

Under exclusion, z reaches y only via x , so $\rho = \beta \pi$.

Hence the **IV estimator** is just the **ratio** of the two:

$$\hat{\beta}_{IV} = \frac{\hat{\rho}}{\hat{\pi}} = \frac{\text{reduced form}}{\text{first stage}}.$$

If the instrument barely moves x ($\hat{\pi} \approx 0$), we divide by ~zero — the estimate explodes. This is the **weak-instrument** problem.

Two-Stage Least Squares (2SLS)

With controls w_i and possibly several instruments, the same idea is implemented in **two stages**:

- **Stage 1 (first stage)**: regress x_i on the instrument(s) and controls, and keep the **fitted values**

$$x_i = \pi z_i + w_i \gamma + \nu_i \longrightarrow \hat{x}_i.$$

$\hat{x}_i$ is the part of x driven *only* by the instrument — purged of U .

- **Stage 2 (second stage)**: regress y_i on $\hat{x}_i$ (not x_i !) and controls:

$$y_i = \beta \hat{x}_i + w_i \delta + \xi_i.$$

With a single instrument this reproduces exactly $\hat{\beta}_{IV} = \hat{\rho}/\hat{\pi}$. In matrix form, with the projection $P_Z = Z(Z'Z)^{-1}Z'$ onto the instruments,

$$\hat{\beta}_{2SLS} = (X'P_ZX)^{-1}X'P_Zy \quad (\text{stage 1 forms } \hat{X} = P_ZX).$$

In practice, estimate 2SLS with a dedicated IV routine (Stata `ivreghdfe`; Python `feols` / `pyfixest`).

What does IV estimate? LATE and compliers

With **heterogeneous** effects β_i , IV recovers not the average effect but a *local* one — under one extra assumption:

Monotonicity (no defiers)

If the instrument affects treatment in one direction — positively (or negatively) — then **no** unit reacts in the opposite direction (no defiers).

- A **complier** is a unit whose treatment the instrument actually moves (treated only when encouraged, untreated otherwise).
- Under relevance, exclusion and monotonicity, the Wald estimator identifies the **Local Average Treatment Effect** — the average effect **for compliers**:

$$\hat{\beta}_{IV} \xrightarrow{p} \mathbb{E}[\beta_i \mid i \text{ complier}] = \text{LATE}.$$

- So IV gives the effect **for compliers**, not the population — different instruments give different LATEs.

Several instruments for one regressor

So far we used a single instrument z for the endogenous x . But we often have **several** valid instruments $z_1, \dots, z_L$ for the *same* x — e.g. for the return to schooling, both distance to college and quarter of birth (defended next).

Excluded vs. included. The z_ℓ are the **excluded instruments**: they appear in the **first stage** (they help predict x) but are **excluded from the outcome equation**. The controls w_i , which appear in *both* equations, are the **included instruments**.

The **exclusion restriction** is exactly the claim that the excluded instruments have **no direct effect** on y — they reach y only through x .

The art of IV: defending each instrument

Example: returns to schooling

x = years of schooling, y = wages. Candidate instruments: z_1 = distance to the nearest college, z_2 = quarter of birth.

For each, argue **relevance** (testable) and **exclusion** (not) — never assumed:

z_1 (**distance to college, Card 1995**). *Relevant?* proximity cuts the cost of attending $\Rightarrow$ more schooling (visible in the first stage). *Excluded?* it reaches wages only through schooling — credible **only after** controls for region, urbanicity and family background (college towns may differ economically).

z_2 (**quarter of birth, Angrist–Krueger 1991**). *Relevant?* school-entry cutoffs and a compulsory-schooling age let those born early leave with less schooling. *Excluded?* birth timing is as-good-as random — but the first stage is **tiny**, a **weak instrument** (Bound–Jaeger–Baker 1995).

The first stage with several instruments

All excluded instruments (together with the controls) enter the **first stage** at once:

First stage

$$x_i = \pi_1 z_{1i} + \cdots + \pi_L z_{Li} + w_i \gamma + \nu_i \longrightarrow \hat{x}_i.$$

- 2SLS forms the fitted value $\hat{x}_i$ — the **single best linear combination** of all the instruments — pooling their information automatically.
- The **second stage is unchanged**: regress y_i on $\hat{x}_i$ (and the controls).
- Relevance is now a **joint** question — do the instruments *together* move x ?

That joint question is what the **first-stage** F (next slide) answers.

Are the instruments strong enough? The first-stage F

First-stage F test

Test the joint null that the excluded instruments do *not* explain the treatment: $H_0 : \pi_1 = \dots = \pi_L = 0$. With a *single* instrument ($L = 1$) it is just the squared first-stage t : $F = t^2$.

- **Weak instruments** (= low F): 2SLS is **biased toward OLS** and its CIs **over-reject** — even with a valid instrument.
- **Rule of thumb**: $F > 10$ (Staiger & Stock, 1997).

Stock–Yogo (2005): the rule of thumb made rigorous

Critical values for the first-stage F to formally reject “weak instruments” at a stated tolerance (e.g. 2SLS bias $\leq 10\%$ of OLS bias); the threshold *rises* with the number of instruments.

- Under heteroskedasticity / clustering, use the **Kleibergen–Paap F** . (F tests **relevance**; the **Hansen J** test, later, tests **validity**.)

CIs that survive weak instruments: Anderson–Rubin

When the first-stage F is low, 2SLS t -based intervals are **unreliable**. The **Anderson–Rubin (AR)** test repairs inference on β .

The idea

To test $H_0 : \beta = \beta_0$, impose it and ask whether the instrument still predicts the imposed residual:

$$y_i - \beta_0 x_i = z_i' \theta + u_i, \quad H_0 : \theta = 0.$$

If β_0 is true, $y_i - \beta_0 x_i$ is exogenous, so z has **no** power.

- The **AR set** = all β_0 not rejected; correct coverage **at any instrument strength**. With one instrument it is **exact**.
- It can be **wider**, unbounded, or empty — itself informative about (non-)identification.
- **Lee, McCrary, Moreira & Porter (2022)**: $F > 10$ is too lenient for honest 5% inference — prefer **AR** (or their tF correction).

Section 2

A detour: Generalized Method of Moments

What is a “moment”?

Step back: *what kind* of estimator is IV? It is a **method of moments**. A **moment** is the expectation of a function of the data — the mean $\mathbb{E}[x]$, a covariance $\mathbb{E}[xy]$, etc.

The method of moments, in one line

A model implies that certain moments take known values. Replace the population expectation $\mathbb{E}[\cdot]$ by its **sample average** $\frac{1}{n} \sum_i$, and solve for the parameters.

IV is exactly this. Exogeneity is a population **moment condition**:

$$\mathbb{E}[z_i \varepsilon_i(\beta)] = 0, \quad \varepsilon_i(\beta) = y_i - \beta x_i$$

(“the instrument is uncorrelated with the error”). We estimate β by making the **sample** version hold:

$$\frac{1}{n} \sum_i z_i (y_i - \beta x_i) = 0 \implies \hat{\beta}_{IV} = \frac{\text{Cov}(z, y)}{\text{Cov}(z, x)}.$$

One equation, one unknown β — solved **exactly**

One instrument, or several?

One instrument (just-identified): one moment, one unknown $\Rightarrow$ exact solution (the case above).

Two instruments for the *same* β . In our schooling example (z_1 = distance to college, z_2 = quarter of birth), each is its own moment condition:

$$\mathbb{E}[z_{1i} \varepsilon_i] = 0 \quad \text{and} \quad \mathbb{E}[z_{2i} \varepsilon_i] = 0.$$

- Each *alone* gives $\hat{\beta}_1 = \frac{\text{Cov}(z_1, y)}{\text{Cov}(z_1, x)}$, $\hat{\beta}_2 = \frac{\text{Cov}(z_2, y)}{\text{Cov}(z_2, x)}$.
- In finite samples $\hat{\beta}_1 \neq \hat{\beta}_2$: **two equations, one unknown — over-identification** ($L = 2 > 1$).

$\rightarrow$ **GMM** answers both: **(1)** how to *combine* them? **(2)** do they *agree* (a test of validity)?

The GMM estimator

Stack the L sample moments,

$$\bar{g}(\beta) = \frac{1}{n} \sum_i z_i (y_i - \beta x_i),$$

and choose β to make them **jointly as close to zero as possible**, in the metric of a weight matrix W :

$$\hat{\beta}_{GMM} = \arg \min_{\beta} \bar{g}(\beta)' W \bar{g}(\beta)$$

- W decides **how much each moment counts**.
- **2SLS is a special case**: GMM with $W = (Z'Z)^{-1}$ (Z the $n \times L$ instrument matrix) — **remember this, it is the bridge to Goldsmith-Pinkham et al.** *Efficient* (two-step) GMM instead sets $W = \hat{S}^{-1}$, the inverse of the moments' covariance — smallest variance under heteroskedasticity.
- Over-identification also buys a **free specification test** of instrument validity — Hansen's J (next slide).

Are the instruments valid? The over-identification test

Over-identified ($L > \#endog$), valid instruments should **agree** up to noise — disagreement flags an invalid one.

Sargan–Hansen J statistic

$$J = n \bar{g}(\hat{\beta})' \hat{S}^{-1} \bar{g}(\hat{\beta}) \xrightarrow{d} \chi^2_{L-\#endog}, \quad H_0 : \mathbb{E}[z_i \varepsilon_i] = 0.$$

Sargan (homoskedastic) / Hansen (robust); dof = $L - 1$ with one endogenous x .

- **Reject** $\Rightarrow$ an over-identifying restriction fails: an instrument is invalid (or the model is misspecified).
- **Just-identified** ($L = \#endog$) $\Rightarrow$ 0 dof: exclusion **cannot be tested**.
- A **joint**, low-power test: it won't say *which* fails, and a pass is **reassuring, not proof**.

Dynamic panels: the GMM you'll actually meet

The GMM you will really meet is **dynamic panel GMM** — used when a **lagged outcome** sits on the right-hand side:

$$y_{it} = \rho y_{i,t-1} + \beta x_{it} + f_i + u_{it}.$$

- **Within breaks here** (Nickell 1981): sweeping out f_i leaves $y_{i,t-1}$ correlated with the demeaned error — biased, badly for small T .
- **Arellano–Bond (1991)**: first-difference to remove f_i , then instrument $\Delta y_{i,t-1}$ with **deeper lagged levels** $y_{i,t-2}, y_{i,t-3}, \dots$
- **Blundell–Bond (1998)** “system GMM”: also use lagged *differences* as instruments for the levels equation (needs an extra stationarity assumption).

...and why dynamic GMM is questionable

Culture, not default

In the **dynamic** case especially, GMM becomes a **black box**: fragile results, and diagnostics that can mislead.

- **Instruments proliferate** with $T \Rightarrow$ it overfits $y_{i,t-1}$, drags $\hat{\beta}$ toward the biased within estimate, and inflates the Hansen J toward $p \approx 1$ — a meaningless “pass” (Roodman 2009).
- **Researcher forks** (lag depth, collapse, one- vs two-step, Windmeijer SE) make a result easy to *fish*.
- **Weak** when y is persistent: lagged levels barely move the differences.

Shift-share is different: *not* a lag-instrument GMM, but an instrument from observed shares and shocks with an **economic story** for exogeneity. Goldsmith-Pinkham et al. (2020) even show it *is* GMM — but a **transparent** one whose identifying variation we can see and defend.

Section 3

The Shift-Share (Bartik) Instrument

Where do good instruments come from?

The binding constraint is almost never *relevance* — it is *exclusion*.

- A valid instrument needs a credible story for why it touches Y **only** through D .
- Natural experiments (lotteries, policy discontinuities, weather...) are scarce.

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- Natural experiments (lotteries, policy discontinuities, weather...) are scarce.

A recurring idea in applied work: many outcomes at the level of a *region, city, or firm* respond to **national or global shocks**, but each unit is exposed to those shocks **differently**, according to its **pre-existing composition**.

That structure — common shocks $\times$ heterogeneous exposure — is exactly the **shift-share** instrument. It is the workhorse of labor, trade, macro, public, and urban economics.

Motivating setting: local labor markets

Classic question (Bartik 1991): what is the elasticity of local **wages** y_i to local **labor demand** x_i across regions i ?

- OLS is biased: a local labor-supply shock (e.g. immigration) moves both employment and wages — $\text{Cov}(x_i, \varepsilon_i) \neq 0$.
- We need an instrument that shifts **local labor demand** for reasons unrelated to local supply.

Idea: a region's industries grow or shrink with *national* industry trends; a region tied to booming industries sees demand rise — regardless of its local supply conditions.

Notation

Throughout the shift-share part:

symbol	meaning
i	unit of analysis (region, city, firm)
$k = 1, \dots, K$	sectors / origins / “shocks”
s_{ik}	share: exposure of unit i to k (e.g. industry k ’s employment share), predetermined
g_k	shift: the common shock to k (e.g. national growth of industry k)
x_i, y_i	endogenous treatment and outcome

Shares are typically (not always) non-negative and **sum to one**:

$$\sum_k s_{ik} = 1.$$

Construction of the instrument

The **shift-share** instrument (traditionally the **Bartik** instrument) interacts predetermined shares with common shifts:

$$z_i = \sum_{k=1}^K s_{ik} g_k$$

- *Shares* s_{ik} — measured at a **baseline** period (before the shocks).
- *Shifts* g_k — common to all units (national / global).

z_i is the **predicted** growth of unit i if each of its sectors grew at the national rate. It is then used as an instrument for the endogenous x_i .

A worked mini-example

Two sectors (manufacturing, services), two regions. Baseline shares:

region	manuf.	services
A	0.70	0.30
B	0.20	0.80

National shifts: $g_{\text{manuf}} = -10\%$ (China shock), $g_{\text{serv}} = +5\%$.

$$z_A = 0.7(-10) + 0.3(5) = -5.5\%, \quad z_B = 0.2(-10) + 0.8(5) = +3.0\%.$$

The instrument predicts region **A contracts**, region **B expands** — purely from *national* trends $\times$ *local* exposure, not from local supply shocks.

Three canonical applications

study	unit i	shares s_{ik}	shifts g_k
Bartik (1991)	region	industry employment shares	national industry growth
Card (2001)	city	origin- k migrant share, 1980	national inflow from origin k
Autor et al. (2013)	region	industry employment shares	Chinese imports (other high-income countries)

Same skeleton everywhere: **predetermined exposure** $\times$ **common shock**. The designs differ in *which component* carries the identifying assumption — the heart of this lecture.

Leave-one-out shifts

A subtle but crucial point: if the shift g_k is built from a national aggregate that **includes unit i itself**, then z_i mechanically contains i 's own outcome shock $\Rightarrow$ correlation with ε_i .

Fix: construct **leave-one-out** shifts that exclude unit i (or, as in Autor et al., measure the shock in *other countries* entirely):

$$g_k^{(-i)} = \text{growth of sector } k \text{ excluding unit } i.$$

This breaks the mechanical link and is standard practice.

The central question

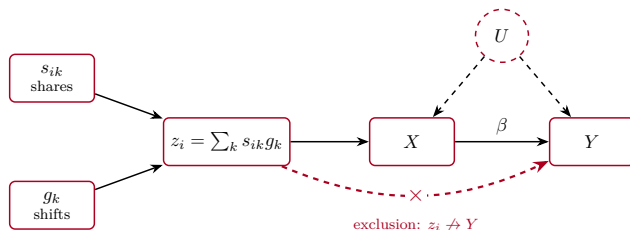
Why is z_i a valid instrument?

$z_i = \sum_k s_{ik} g_k$ has **two** ingredients. Exogeneity can be argued to come from *either* one:

- **From the shares s_{ik}** — units differ in *exposure*, and that exposure is as-good-as-random (conditional on controls). → **Goldsmith-Pinkham et al. (2020)**.
- **From the shifts g_k** — the *shocks* are as-good-as-random, even if exposure is endogenous. → **Borusyak et al. (2022)**.

These are **different research designs** with different assumptions, diagnostics, and standard errors. We take each in turn.

The two routes, in one picture



Both ingredients feed z_i ; its exogeneity (clean of U) can be earned from **either**:

- **Goldsmith-Pinkham et al. (shares):** the shares are as-good-as-random $\Rightarrow \mathbb{E}[s_{ik}\varepsilon_i] = 0$.
- **Borusyak et al. (shifts):** the shifts are as-good-as-random $\Rightarrow$ valid *even if shares are endogenous*.

Section 4

Identification “from the shares”:
Goldsmith-Pinkham et al. (2020)

The big idea

Goldsmith-Pinkham et al. (2020) ask: *what is a Bartik regression actually doing?*

Numerical equivalence (their Proposition 1.1)

The 2SLS estimator using the single instrument $z_i = \sum_k s_{ik}g_k$ is *numerically identical* to a GMM estimator that uses the K industry shares $\{s_{ik}\}_k$ as **separate instruments**, with a weight matrix built from the shifts.

Formally, with z the instrument, x the treatment, y the outcome (stacked over units), S the matrix of shares and W a weight matrix:

$$\hat{\beta}_{Bartik} = \frac{z'y}{z'x}, \quad \hat{\beta}_{GMM} = \frac{x'SWS'y}{x'SWS'x}.$$

With the rank-one weight matrix $W = gg'$ (the outer product of the K -vector of shifts g) and $z = Sg$ (the $n \times K$ shares matrix S times g), the two coincide: $Sgg'S' = (Sg)(Sg)' = zz'$, so $\hat{\beta}_{GMM} = \hat{\beta}_{Bartik}$.

Why this reframes identification

If Bartik = “shares as instruments”, then the **exclusion restriction lives on the shares**:

$$\mathbb{E}[s_{ik} \varepsilon_i] = 0 \quad \text{for every } k.$$

The shifts g_k are *not* instruments here — they only **combine** the K share-instruments into one. The identifying content is: *is each exposure share uncorrelated with the unobservable?*

This is the natural reading when you have **few, dominant sectors** and tell a story like “places more exposed to industry k form a clean comparison group.”

Opening the black box: Rotemberg weights

Building on Rotemberg (1983), Goldsmith-Pinkham et al. decompose the over-identified estimator into the K **just-identified** estimators, one per share-instrument:

Proposition 3.1 (Goldsmith-Pinkham et al.)

$$\hat{\beta}_{Bartik} = \sum_{k=1}^K \hat{\alpha}_k \hat{\beta}_k, \quad \sum_k \hat{\alpha}_k = 1,$$

where the just-identified estimate and the **Rotemberg weight** are

$$\hat{\beta}_k = (s'_k x)^{-1} s'_k y, \quad \hat{\alpha}_k = \frac{g_k s'_k x}{\sum_{k'} g_{k'} s'_{k'} x}.$$

(Here x, y are residualized on the controls; $s_k = \{s_{ik}\}_i$, the k -th share column.)

So Bartik is a **weighted average of single-share IV estimates**.

Reading the Rotemberg weights

The weight $\hat{\alpha}_k$ tells you **how much industry k drives the headline estimate** — and how sensitive $\hat{\beta}$ is to a violation of exogeneity in that one share. Practical lessons from Goldsmith-Pinkham et al.:

- Weights measure **sensitivity to misspecification** (Andrews–Gentzkow–Shapiro 2017): bias in a high- $\hat{\alpha}_k$ share passes strongly into $\hat{\beta}$.
- The weights **sum to one but can be negative** — an *affine*, not convex, average; a warning for the LATE interpretation under heterogeneous effects.
- The **shifts g_k explain little** of the weights ($< 1\%$ in their example) — “the variation comes from the shocks” is often *wrong*.
- In their application, **5 industries hold $> 40\%$ of the weight** (oil & gas on top).

Lineage: that IV recovers a **weighted average of heterogeneous effects** dates to Imbens & Angrist (1994) / Angrist & Imbens (1995); negative weights under many instruments are sharpened by Mogstad, Torgovitsky & Walters (2021).

The identifying assumption (shares view)

Share exogeneity

The exposure shares s_{ik} are uncorrelated with the unit-level unobservable ε_i (conditional on controls).

When outcomes are in changes, this is a **parallel-trends** condition, exactly as in difference-in-differences: absent the shock, exposed and unexposed units would have trended alike. The shift-share pools **many DiD comparisons**, one per share. Inference here relies on **many units** (large N).

Diagnostics (do these for the high-weight shares):

- correlate top-weight shares with observable confounders / baseline characteristics;
- test for **pre-trends** in the top-weight share comparisons and in z_i overall;
- report **over-identification** tests and the spread of the $\hat{\beta}_k$;
- use the **bartik weight** package to compute the weights.

Section 5

Identification “from the shifts”: Borusyak et al. (2022)

The big idea

Borusyak et al. (2022) start from a different intuition:

A share-weighted average of random shifts is itself as-good-as-random

Even if the shares s_{ik} are *endogenous* (exposed units differ systematically), as long as the shocks g_k are as-good-as-randomly assigned across many sectors, the instrument $z_i = \sum_k s_{ik}g_k$ is mean-independent of ε_i on average.

Example: skill-intensive regions may attract particular immigrants (endogenous shares), but if subsidies are randomized across many high- and low-skill industries, exposed regions get **typical** instrument values.

The shift-level “translation”

Borusyak et al. show the shift-share IV is a **translation device**: the location-level IV estimator is numerically equal to a **shock-level IV regression** that uses g_k *directly* as the instrument.

- Aggregate the outcome and treatment to the **shock level** using exposure weights $\omega_k \propto \sum_i e_i s_{ik}$, **normalized** so $\sum_k \omega_k = 1$ (e_i a unit weight).
- Run IV of the exposure-weighted $\bar{y}_k$ on $\bar{x}_k$, instrumenting with g_k .

The **same number** comes out — but now the assumption is stated on the **shocks**, and there are as many “observations” as there are **shocks** K , not units.

How many shifts do you really have?

Because each shift carries an importance weight ω_k , the **effective sample size** is governed by the concentration of those weights:

Effective number of shifts

$$K_{\text{eff}} = \frac{1}{\sum_k \omega_k^2} = \frac{1}{\text{HHI of exposure weights}}.$$

- If a few sectors dominate exposure, K_{eff} is **small** — the design rests on a handful of shocks and is fragile.
- The shifts approach needs K_{eff} **large** so a law of large numbers kicks in.

The identifying assumptions (shifts view)

Two requirements

- (1) **Quasi-random shocks (shock orthogonality):** the g_k are as-good-as randomly assigned and affect y only through x — mean-independent of the exposure-weighted unobservable.
- (2) **Many, dispersed shocks:** K_{eff} large (low exposure Herfindahl), shocks mutually uncorrelated.

Notably, **no assumption on the shares** is required — they may be endogenous. This fits “generic” industry shares (as in Autor et al. 2013) where share exogeneity would be implausible.

Controls and diagnostics (shifts view)

Make shocks conditionally as-good-as-random with the right controls:

- If shares do **not** sum to one, **control for the sum of shares** $\sum_k s_{ik}$.
- For a shock-level confounder q_k (e.g. skill intensity), control for its **shift-share aggregate** $\sum_k s_{ik}q_k$ — same shares, applied to q_k .

Diagnostics:

- **Balance tests** at the shock level: regress g_k on shock-level characteristics.
- Report K_{eff} and exposure-weighted summary statistics.
- Use leave-one-out / externally-measured shifts to kill mechanical correlation.
- The **ssaggregate** package builds the shock-level dataset.

Standard errors also need special care under shared exposure — the next section.

Section 6

Inference

Why standard errors are wrong

A subtle problem (Adão et al. 2019): **two regions with similar exposure share vectors get similar instrument values z_i** , because z_i is the *same* linear combination of common shocks.

⇒ regression residuals are **correlated across units that look unrelated** (even in different states/clusters).

Conventional, heteroskedasticity-robust, and even region-clustered standard errors **ignore this** and badly **over-reject** — a nominal 5% test can reject 30–50% of the time.

The fix: exposure-robust standard errors

Practical guidance

Use standard errors that are robust to cross-unit correlation *induced by shared exposure*:

- **Exposure-robust** estimators of Adão et al. (2019) — valid under the exposure structure;
- equivalently, **shock-level (Borusyak et al.) standard errors** from the shift-level IV regression cluster naturally **at the level of the shocks k** ;
- both are implemented in the authors' Stata/R packages.

Rule of thumb: the unit of “independent variation” is the **shock**, not the region — cluster your thinking (and SEs) accordingly.

Section 7

Shares or shifts? A practical guide

Two paths, one instrument

	Goldsmith-Pinkham et al.	Borusyak et al.
Exogenous	the shares s_{ik}	the shifts g_k
Shares may be endogenous?	no	yes
Asymptotics in	number of units	number of shocks K
Best when	few, dominant sectors; “tailored” shares	many small shocks; “generic” shares
Key diagnostic	Rotemberg weights, pre-trends	balance of g_k , K_{eff}
Inference	over-id tests; exposure-robust SE	shock-level / Adão et al. SE
Canonical	Card (2001) migration enclaves	Autor et al. (2013) China shock

Which assumption fits your design?

Ask yourself:

- ❶ **Can I defend a few shares as exogenous?** Are they *tailored* (they mediate only the shock of interest, like origin-specific migrant shares)? → **shares view**.
- ❷ **Are my shares “generic”** (industry composition that proxies exposure to *many* unobserved shocks)? Then share exogeneity is implausible → lean on **exogenous shifts**.
- ❸ **Do I have many, plausibly-random shocks** with low exposure HHI? → **shifts view** is available and powerful.

In practice: report **both** sets of diagnostics, and always use **exposure-robust standard errors**.

Section 8

Applied session

Estimating a Bartik design in Python

```
import pyfixest as pf

# 1. Instrument:  $z_i = \sum_k s_{ik} * g_k$  (leave-one-out  $g$ )
df["z"] = (shares * shifts).sum(axis=1)

# 2. First stage - read its  $F$  (rule of thumb  $> 10$ )
fs = pf.feols("x ~ z", data=df); fs.summary()

# 3. 2SLS (region clustering is NOT valid here; see
# Inference - use exposure-robust / shock-level SEs)
iv = pf.feols("y ~ 1 | x ~ z", data=df); iv.summary()
```

- Read the **first-stage** F (recall Lee et al.: prefer **AR** when weak).
- **Region clustering over-rejects** — use shock-level (ssaggregate) / **exposure-robust** SEs (Adão et al.), and inspect **Rotemberg weights** (bartik_weight).

- Shift-share = **common shocks** $\times$ **predetermined exposure**: a way to manufacture an instrument where none is obvious.
- The instrument has **two ingredients**; identification can rest on **either**:
 - **shares** exogenous (Goldsmith-Pinkham et al.) — pools many DiD comparisons; read the **Rotemberg weights**;
 - **shifts** exogenous (Borusyak et al.) — many quasi-random shocks; mind K_{eff} .
- **Inference is special**: shared exposure correlates residuals — use **exposure-robust standard errors** (Adão et al. / shock-level).
- Match the **story** to the **assumption**, and **diagnose** the one you rely on.

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- Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. “Bartik Instruments: What, When, Why, and How.” *American Economic Review* 110 (8): 2586–624. <https://doi.org/10.1257/aer.20181047>.